

task 5.

8.5.19

$$\int \frac{dx}{\cos x} = \left[\begin{array}{l} \text{tr.} \\ t = \tan \frac{x}{2} \Rightarrow x = 2 \arctan t \Rightarrow dx = \frac{2 dt}{1+t^2}, \\ \cos x = \frac{1-t^2}{1+t^2} \end{array} \right] =$$

$$= \int \frac{\frac{2 dt}{1+t^2}}{\frac{1-t^2}{1+t^2}} = \int \frac{2 dt \cdot (1+t^2)}{(1+t^2) \cdot (1-t^2)} = \int \frac{2 dt}{1-t^2} =$$

$$= -2 \int \frac{dt}{t^2-1} = [\text{rad.}] = -2 \cdot \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| + C =$$

$$= -\ln \left| \frac{\tan \frac{x}{2} - 1}{\tan \frac{x}{2} + 1} \right| + C$$

8.5.20

$$\int \frac{dx}{1-\sin x} = \left[\begin{array}{l} \text{tr.} \\ t = \tan \frac{x}{2} \Rightarrow x = 2 \arctan t \Rightarrow dx = \frac{2 dt}{1+t^2}, \\ \sin x = \frac{2t}{1+t^2} \end{array} \right] =$$

$$= \int \frac{\frac{2 dt}{1+t^2}}{1 - \frac{2t}{1+t^2}} = \int \frac{2 dt \cdot (1+t^2)}{(1+t^2) \cdot (1+t^2 - 2t)} = \int \frac{2 dt}{-2t + 1 + t^2} =$$

$$= \int \frac{2 dt}{(t-1)^2} = 2 \int \frac{dt}{(t-1)^2} = 2 \int \frac{d(t-1)}{(t-1)^2} = -\frac{2}{t-1} + C =$$

$$= -\frac{2}{\tan \frac{x}{2} - 1} + C$$

8.5.21

$$\int \frac{dx}{5+4\sin x} = \int \frac{dx}{4\sin x + 5} = \left[\begin{array}{l} \text{tr.} \\ t = \tan \frac{x}{2} \Rightarrow x = 2 \arctan t \Rightarrow \\ dx = \frac{2 dt}{1+t^2}, \\ \sin x = \frac{2t}{1+t^2} \end{array} \right] =$$

$$\begin{aligned}
 &= \int \frac{\frac{2dt}{1+t^2}}{4 \cdot \frac{2t}{1+t^2} + 5} = \int \frac{\frac{2dt}{1+t^2}}{\frac{8t + 5(1+t^2)}{1+t^2}} = \int \frac{2dt \cdot (1+t^2)}{(1+t^2) \cdot (8t + 5 + 5t^2)} = \\
 &= \int \frac{2dt}{5t^2 + 8t + 5} = \frac{2}{5} \int \frac{dt}{t^2 + \frac{8}{5}t + 1} = \frac{2}{5} \int \frac{dt}{t^2 + \frac{8}{5}t + \frac{16}{25} + \frac{9}{25}} = \\
 &= \frac{2}{5} \int \frac{dt}{\left(t + \frac{4}{5}\right)^2 + \frac{9}{25}} = \frac{2}{5} \int \frac{d\left(t + \frac{4}{5}\right)}{\left(t + \frac{4}{5}\right)^2 + \frac{9}{25}} = \frac{2}{5} \int \frac{d\left(t + \frac{4}{5}\right)}{\left(t + \frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \\
 &= \frac{2}{5} \cdot \frac{5}{3} \cdot \operatorname{arctg} \frac{5\left(t + \frac{4}{5}\right)}{3} + C = \frac{2}{3} \operatorname{arctg} \frac{5t+4}{3} + C = \\
 &= \frac{2}{3} \operatorname{arctg} \frac{5\operatorname{tg} \frac{x}{2} + 4}{3} + C
 \end{aligned}$$

8.5.22

$$\begin{aligned}
 &\int \frac{2 - \sin x}{2 + \cos x} dx = \int \left(\frac{2}{2 + \cos x} - \frac{\sin x}{2 + \cos x} \right) dx = \left[\begin{array}{l} u = 2 + \cos x \\ u' = -\sin x \end{array} \right] = \\
 &= 2 \int \frac{1}{2 + \frac{1 - \operatorname{tg}^2 \frac{x}{2}}{1 + \operatorname{tg}^2 \frac{x}{2}}} dx - \int -\frac{1}{u} du = \left[t = \operatorname{tg} \frac{x}{2} \quad t' = \frac{1}{2} (1 + \operatorname{tg}^2 \frac{x}{2}) \right] = \\
 &= 2 \int \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{1}{\frac{1}{2}(1+t^2)} dt + \ln|u| + C = \\
 &= 2 \int \frac{1}{\frac{2+2t^2+1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt + \ln|u| + C = \\
 &= 2 \int \frac{1}{\frac{t^2+3}{1+t^2}} \cdot \frac{2}{1+t^2} dt + \ln|u| + C =
 \end{aligned}$$

$$= 2 \int \frac{1+t^2}{t^2+3} \cdot \frac{2}{1+t^2} dt + \ln|u| + C =$$

$$= 4 \int \frac{dt}{t^2+3} + \ln|2+\cos x| + C =$$

$$= 4 \frac{1}{\sqrt{3}} \operatorname{arctg} \frac{t}{\sqrt{3}} + \ln|2+\cos x| + C =$$

$$= \frac{4}{\sqrt{3}} \operatorname{arctg} \frac{\operatorname{tg} \frac{x}{2}}{\sqrt{3}} + \ln|2+\cos x| + C =$$

$$= \frac{4\sqrt{3}}{3} \operatorname{arctg} \frac{\sqrt{3} \operatorname{tg} \frac{x}{2}}{3} + \ln|2+\cos x| + C.$$

8.5.23

$$\int \frac{dx}{2 \sin x - \cos x + 5} = \left[\begin{array}{l} \text{pen. } t = \operatorname{tg} \frac{x}{2} \Rightarrow x = 2 \operatorname{arctg} t = \\ dx = \frac{2dt}{1+t^2}, \quad \sin x = \frac{2t}{1+t^2}, \\ \cos x = \frac{1-t^2}{1+t^2} \end{array} \right] =$$

$$= \int \frac{\frac{2dt}{1+t^2}}{2 \cdot \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} + 5} =$$

$$= \int \frac{2dt \cdot (1+t^2)}{(1+t^2) \cdot (4t - 1 + t^2 + 5 + 5t^2)} =$$

$$= \int \frac{2dt}{6t^2 + 4t + 4} = \int \frac{dt}{3t^2 + 2t + 2} = \frac{1}{3} \int \frac{dt}{t^2 + \frac{2}{3}t + \frac{2}{3}} =$$

$$= \frac{1}{3} \int \frac{dt}{t^2 + \frac{2}{3}t + \frac{1}{9} + \frac{5}{9}} = \frac{1}{3} \int \frac{dt}{(t + \frac{1}{3})^2 + \frac{5}{9}} =$$

$$= \frac{1}{3} \int \frac{d(t + \frac{1}{3})}{(t + \frac{1}{3})^2 + \frac{5}{9}} = \frac{1}{3} \cdot \frac{3}{\sqrt{5}} \operatorname{arctg} \frac{(t + \frac{1}{3}) \cdot 3}{\sqrt{5}} =$$

$$= \frac{1}{\sqrt{5}} \operatorname{arctg} \frac{3t + 1}{\sqrt{5}} = \frac{\sqrt{5}}{5} \operatorname{arctg} \frac{3 \operatorname{tg} \frac{x}{2} + 1}{\sqrt{5}}$$

8.5.24

$$\begin{aligned} \int \frac{1 + \sin x}{(1 + \cos x) \sin x} dx &= \int \frac{dx}{(1 + \cos x) \sin x} + \int \frac{\sin x dx}{(1 + \cos x) \sin x} = \\ &= \int \frac{dx}{\sin x + \cos x \sin x} + \int \frac{dx}{1 + \cos x} = \left[\begin{array}{l} t = \operatorname{tg} \frac{x}{2} \\ t' = \frac{1}{2}(1 + t^2) \end{array} \right] = \\ &= \int \frac{2}{\left(\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} \cdot \frac{2t}{1+t^2} \right) (1+t^2)} dx + \int \frac{1}{1 + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt = \\ &= \int \frac{2t}{1+t^2} + \frac{2t(1-t^2)}{(1+t^2)^2} \cdot \frac{2}{1+t^2} dt + \int \frac{1}{1+t^2+1-t^2} \cdot \frac{2}{1+t^2} dt = \\ &= \int \frac{2t + 2t^3 + 2t - 2t^3}{(1+t^2)^2} \cdot \frac{2}{1+t^2} dt + \int \frac{1+t^2}{2} \cdot \frac{2}{1+t^2} dt = \\ &= \int \frac{1+t^2}{2t} dt + \int dt = \frac{1}{2} \int \frac{dt}{t} + \frac{1}{2} \int t dt + \int dt = \\ &= \frac{1}{2} \ln|t| + \frac{1}{2} \cdot \frac{t^2}{2} + t + C = \\ &= \frac{1}{2} \ln \left| \operatorname{tg} \frac{x}{2} \right| + \frac{\operatorname{tg}^2 \frac{x}{2}}{4} + \operatorname{tg} \frac{x}{2} + C. \end{aligned}$$

8.5.25

$$\int \frac{dx}{5 \sin^2 x - 3 \cos^2 x + 4} = \int \frac{1}{5(-\sin x)^2 - 3(-\cos x)^2 + 4} = \int \frac{1}{5 \sin^2 x - 3 \cos^2 x + 4} = R(\sin x, \cos x)$$

$\Rightarrow$ enge 2 $\Rightarrow t = \operatorname{tg} x \Rightarrow dx = \frac{dt}{1+t^2}$, $\sin x = \frac{t}{\sqrt{1+t^2}}$, $\cos x = \frac{1}{\sqrt{1+t^2}}$] =

$$= \int \frac{\frac{dt}{1+t^2}}{5\left(\frac{t}{\sqrt{1+t^2}}\right)^2 - 3\left(\frac{1}{\sqrt{1+t^2}}\right)^2 + 4} =$$

$$= \int \frac{dt \cdot (1+t^2)}{(1+t^2) \cdot (5t^2 - 3 + 4 + 4t^2)} = \int \frac{dt}{9t^2 + 1} = \frac{1}{9} \int \frac{dt}{t^2 + \frac{1}{9}} =$$

$$\cancel{\frac{1}{9}} = \frac{1}{9} \cdot \operatorname{arctg}(3t) + C =$$

$$= \frac{1}{3} \operatorname{arctg}(3 \operatorname{tg} x) + C$$

8.5.26.

$$\int \frac{dx}{4\sin^2 x + 9\cos^2 x} = \left[\text{cn. 2. } \Rightarrow t = \operatorname{tg} x \Rightarrow dx = \frac{dt}{1+t^2}, \right. \\ \left. \sin x = \frac{t}{\sqrt{1+t^2}}, \cos x = \frac{1}{\sqrt{1+t^2}} \right] =$$

$$= \int \frac{\frac{dt}{1+t^2}}{4\left(\frac{t}{\sqrt{1+t^2}}\right)^2 + 9\left(\frac{1}{\sqrt{1+t^2}}\right)^2} =$$

$$= \int \frac{dt \cdot (1+t^2)}{(1+t^2) \cdot (4t^2 + 9)} = \frac{1}{4} \int \frac{dt}{t^2 + \frac{9}{4}} = \frac{1}{4} \int \frac{dt}{t^2 + \left(\frac{3}{2}\right)^2} =$$

$$= \frac{1}{4} \cdot \frac{2}{3} \operatorname{arctg} \frac{2t}{3} + C =$$

$$= \frac{1}{6} \operatorname{arctg} \frac{2 \operatorname{tg} x}{3} + C$$

8.5.27

$$\int \frac{dx}{1+3\cos^2 x} = \left[\text{cn. 2. } \Rightarrow t = \operatorname{tg} x \Rightarrow dx = \frac{dt}{1+t^2}, \right. \\ \left. \cos x = \frac{1}{\sqrt{1+t^2}} \right] =$$

$$= \int \frac{\frac{dt}{1+t^2}}{1 + 3\left(\frac{t}{\sqrt{1+t^2}}\right)^2} = \int \frac{dt \cdot (1+t^2)}{(1+t^2) \cdot (1+t^2+3)} =$$

$$= \int \frac{dt}{t^2+4} = \frac{1}{2} \operatorname{arctg} \frac{t}{2} + C = \frac{1}{2} \operatorname{arctg} \frac{\operatorname{tg} x}{2} + C$$

8.5.28

$$\int \frac{dx}{\sin^4 x} = \int R(-\sin x) = \frac{1}{(\sin x)^4} = \frac{1}{\sin^4 x} = R(\sin x) \Rightarrow \text{cn. 2.} \Rightarrow$$

$$\Rightarrow t = \operatorname{tg} x \Rightarrow dx = \frac{dt}{1+t^2}, \quad \sin x = \frac{t}{\sqrt{1+t^2}} \quad \int =$$

$$= \int \frac{\frac{dt}{1+t^2}}{\left(\frac{t}{\sqrt{1+t^2}}\right)^4} = \int \frac{dt \cdot (1+t^2)^2}{(1+t^2) \cdot t^4} = \int \frac{1+t^2}{t^4} dt =$$

$$= \int \frac{dt}{t^4} + \int \frac{dt}{t^2} = \int \frac{dt}{t^4} + \int \frac{dt}{t^2} =$$

$$= \frac{t^{-3}}{-3} + \left(-\frac{1}{t}\right) + C = -\frac{1}{3 \operatorname{tg}^3 x} - \frac{1}{\operatorname{tg} x} + C$$

8.5.29.

$$\begin{aligned} \int \sin^5 x \, dx &= \int (1 - \cos^2 x)^2 \sin x \, dx = \left[t = \cos x \Rightarrow dx = -\frac{1}{\sin x} dt \right] = \\ &= -\int (1-t^2)^2 dt = -\int (1 - 2t^2 + t^4) dt = -\left(\int 1 dt - \int 2t^2 dt + \int t^4 dt\right) = \\ &= -\left(\frac{t^5}{5} - 2\frac{t^3}{3} + t\right) + C = -\frac{1}{5} \cos^5 x + \frac{2}{3} \cos^3 x - \cos x + C. \end{aligned}$$

8.5.50

$$\begin{aligned} \int \sin^4 x \cdot \cos^5 x \, dx &= [\text{cn. 3.2.} \Rightarrow t = \sin x \Rightarrow dx = \frac{dt}{\sqrt{1-t^2}}, \\ \cos x = \sqrt{1-t^2}] &= \int t^4 \cdot (\sqrt{1-t^2})^5 \cdot \frac{dt}{\sqrt{1-t^2}} = \int t^4 \cdot (1-t^2)^2 \cdot dt = \end{aligned}$$

$$\begin{aligned}
 &= \int t^4 \cdot (1 - 2t^2 + t^4) dt = \int (t^4 - 2t^6 + t^8) dt = \\
 &= \int t^4 dt - 2 \int t^6 dt + \int t^8 dt = \\
 &= \frac{t^5}{5} - \frac{2t^7}{7} + \frac{t^9}{9} + C = \frac{\sin^5 x}{5} - \frac{2\sin^7 x}{7} + \frac{\sin^9 x}{9} + C.
 \end{aligned}$$

8.5.31

$$\begin{aligned}
 \int \frac{\sin 2x dx}{\cos^5 x} &= \int \frac{2 \sin x \cos x dx}{\cos^5 x} = 2 \int \frac{\sin x}{\cos^4 x} dx = \\
 &= \left[\text{cn. s. t. } t = \cos x \Rightarrow dx = -\frac{dt}{\sqrt{1-t^2}}, \sin x = \sqrt{1-t^2} \right] = \\
 &= 2 \int \frac{\sqrt{1-t^2}}{t^4} \cdot \left(-\frac{dt}{\sqrt{1-t^2}} \right) = -2 \int \frac{dt}{t^4} = -2 \cdot \left(-\frac{1}{5t^5} \right) + C = \\
 &= \frac{2}{5 \cos^5 x} + C.
 \end{aligned}$$

8.5.32

$$\begin{aligned}
 \int \frac{\sin^4 x dx}{\cos x} &= \left[\text{cn. s. z. } t = \sin x \Rightarrow dx = \frac{dt}{\sqrt{1-t^2}}, \right. \\
 &\quad \left. \cos x = \sqrt{1-t^2} \right] = \\
 &= \int \frac{t^4}{\sqrt{1-t^2}} \cdot \frac{dt}{\sqrt{1-t^2}} = \int \frac{t^4 dt}{1-t^2} = - \int \frac{t^4}{t^2-1} dt = \\
 &= - \int \frac{t^4 + 1 + 1}{t^2-1} dt = - \left(\int \frac{(t^2-1)(t^2+1)}{t^2-1} dt + \int \frac{dt}{t^2-1} \right) =
 \end{aligned}$$

$$\begin{aligned}
 &= - \left(\int (t^2 + 1) dt + \int \frac{dt}{t^2 - 1} \right) = - \int t^2 dt - \int dt - \int \frac{dt}{t^2 - 1} = \\
 &= - \frac{t^3}{3} - t - \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| + C = \\
 &= - \frac{\tan^3 x}{3} - \tan x - \frac{1}{2} \ln \left| \frac{\tan x - 1}{\tan x + 1} \right| + C
 \end{aligned}$$

8.5.33

$$\begin{aligned}
 \int \sin^6 x \, dx &= \left[\text{ex 4} \Rightarrow \sin^2 x = \frac{1 - \cos 2x}{2} \right] = \\
 &= \int \left(\frac{1 - \cos 2x}{2} \right)^3 dx = \int \frac{(1 - \cos 2x)^3}{8} dx = \\
 &= \frac{1}{8} \left(\int (1 - 3\cos 2x + 3\cos^2 2x - \cos^3 2x) dx \right) = \\
 &= \frac{1}{8} x - \frac{3}{16} \sin 2x + \frac{3x}{16} + \frac{3 \sin 4x}{64} - \frac{\sin 2x}{16} + \frac{\sin^3 2x}{48} + C = \\
 &= \frac{5x}{16} - \frac{\sin 2x}{4} + \frac{\sin^3 2x}{48} + \frac{3 \sin 4x}{64} + C
 \end{aligned}$$

8.5.34

$$\begin{aligned}
 \int \sin^2 x \cos^4 x \, dx &= \int \frac{(1 + \cos 2x)^2}{2} \cdot \frac{1 - \cos 2x}{4} dx = \\
 &= \int \frac{(1 + \cos 2x)(1 - \cos 2x)^2}{8} dx = \left[u = 2x \rightarrow u' = 2 \Rightarrow dx = \frac{1}{2} du \right] = \\
 &= \frac{1}{16} \int (1 + \cos u)(1 - \cos u)^2 du = \frac{1}{16} \int (\cos^3 u - \cos^2 u -
 \end{aligned}$$

$$\begin{aligned}
 & -\cos u + 1) du = \frac{\sin u}{16} - \frac{\sin^3 u}{48} - \frac{u}{32} - \frac{\sin 2u}{64} - \frac{\sin 4u}{16} + \frac{u}{16} + C \\
 & = \frac{u}{32} - \frac{\sin 2u}{64} - \frac{\sin^3 u}{48} + C = \\
 & = \frac{x}{16} - \frac{\sin 4x}{64} - \frac{\sin^3 2x}{48} + C.
 \end{aligned}$$

8.5.36.

$$\begin{aligned}
 \int \sin x \cdot \sin 3x dx &= \int \frac{1}{2} (\cos(-2x) - \cos 4x) dx = \\
 &= \frac{1}{2} \int (\cos 2x - \cos 4x) dx = \frac{1}{2} \left(\int \cos 2x dx - \int \cos 4x dx \right) = \\
 &= \frac{1}{2} \cdot \left(\frac{\sin 2x}{2} - \frac{\sin 4x}{4} \right) + C = \frac{\sin 2x}{4} - \frac{\sin 4x}{8} + C
 \end{aligned}$$

8.5.37

$$\begin{aligned}
 \int \sin \frac{x}{12} \cdot \cos \frac{x}{3} dx &= \int \frac{1}{2} \left(\sin \frac{5x}{12} + \sin \left(-\frac{x}{4} \right) \right) dx = \\
 &= \frac{1}{2} \left(\int \sin \frac{5x}{12} dx - \int \sin \frac{x}{4} dx \right) = \\
 &= \frac{1}{2} \left(-\frac{12}{5} \cos \frac{5x}{12} + 4 \cos \frac{x}{4} \right) + C = -\frac{6}{5} \cos \frac{5x}{12} + 2 \cos \frac{x}{4} + C
 \end{aligned}$$

8.5.38

$$\begin{aligned}
 \int \cos x \cdot \cos 3x dx &= \int \frac{1}{2} (\cos(-2x) + \cos 4x) dx = \\
 &= \frac{1}{2} \left(\int \cos 2x dx + \int \cos 4x dx \right) = \\
 &= \frac{1}{2} \cdot \left(\frac{\sin 2x}{2} + \frac{\sin 4x}{4} \right) + C = \\
 &= \frac{\sin 2x}{4} + \frac{\sin 4x}{8} + C.
 \end{aligned}$$

$$\begin{aligned}
 8.5.39 \quad \int \cos x \cdot \cos 3x \cdot \cos 5x \, dx &= \int \frac{1}{2} (\cos(-2x) + \cos 4x) \cdot \cos 5x \, dx = \\
 &= \frac{1}{2} \int \cos 5x \cdot (\cos 2x + \cos 4x) \, dx = \\
 &= \frac{1}{2} \int (\cos 5x \cos 2x + \cos 5x \cos 4x) \, dx = \\
 &= \frac{1}{2} \left(\int \cos 5x \cos 2x \, dx + \int \cos 5x \cos 4x \, dx \right) = \\
 &= \frac{1}{2} \left(\frac{1}{2} \int (\cos 3x + \cos 7x) \, dx + \frac{1}{2} \int (\cos x + \cos 9x) \, dx \right) = \\
 &= \frac{1}{4} \left(\int \cos 3x \, dx + \int \cos 7x \, dx + \int \cos x \, dx + \int \cos 9x \, dx \right) = \\
 &= \frac{1}{4} \left(\frac{\sin 3x}{3} + \frac{\sin 7x}{7} + \sin x + \frac{\sin 9x}{9} \right) + C = \\
 &= \frac{\sin 3x}{12} + \frac{\sin 7x}{28} + \frac{\sin x}{4} + \frac{\sin 9x}{36} + C.
 \end{aligned}$$

8.5.40

$$\begin{aligned}
 &\int \cot^4 x \, dx = \int (\csc^2 x - 1) \cot^2 x \, dx = \\
 &= \int \cot^2 x \csc^2 x \, dx = \int \cot^2 x \cdot \left(\frac{1}{\sin^2 x} - 1 \right) \, dx = \\
 &= \int \cot^2 x \cdot \frac{dx}{\sin^2 x} - \int \cot^2 x \, dx = \left[t = \cot x \rightarrow dt = -\frac{dx}{\sin^2 x} \right] = \\
 &= \int t^2 dt - \int \cot^2 x \cdot \cot^2 x \, dx = \int t^2 dt - \int \cot^2 x \cdot \left(\frac{1}{\sin^2 x} - 1 \right) \, dx = \\
 &= \int t^2 dt - \int \cot^2 x \cdot \frac{dx}{\sin^2 x} - \int \cot^2 x \, dx = \int t^2 dt - \int t^2 dt - \int \left(\frac{1}{\sin^2 x} - 1 \right) \, dx = \\
 &= \int t^2 dt - \int t^2 dt - \int \frac{dx}{\sin^2 x} - \int dx =
 \end{aligned}$$

$$= \frac{t^5}{5} - \frac{t^3}{3} + \operatorname{ctg} x - x + C =$$

$$= \frac{\operatorname{ctg}^5 x}{5} - \frac{\operatorname{ctg}^3 x}{3} + \operatorname{ctg} x - x + C.$$

8.5.41 $\int \operatorname{tg}^4 \frac{x}{2} dx = [t = \frac{x}{2} \Rightarrow t' = \frac{1}{2} \Rightarrow dx = 2dt] =$

$$= 2 \int \operatorname{tg}^4 t dt = [c.n. 6. \operatorname{tg}^2 x = \frac{1}{\cos^2 x} - 1] =$$

$$= 2 \int \operatorname{tg}^2 t \cdot (\frac{1}{\cos^2 t} - 1) dt = 2 \left(\int \operatorname{tg}^2 t \cdot \frac{dt}{\cos^2 t} - \int \operatorname{tg}^2 t dt \right) =$$

$$= [u = \operatorname{tg} x \Rightarrow du = \frac{dx}{\cos^2 x}] = 2 \left(\int u^2 du - \int (\frac{1}{\cos^2 t} - 1) dt \right) =$$

$$= 2 \left(\int u^2 du - \int \frac{dt}{\cos^2 t} - \int dt \right) =$$

$$= 2 \left(\frac{u^3}{3} - \operatorname{tg} t - t \right) + C =$$

$$= 2 \left(\frac{\operatorname{tg}^3 t}{3} - \operatorname{tg} t - t \right) + C =$$

$$= \frac{2 \operatorname{tg}^3 \frac{x}{2}}{3} - 2 \operatorname{tg} \frac{x}{2} - \frac{2x}{2} + C = \frac{2 \operatorname{tg}^3 \frac{x}{2}}{3} - 2 \operatorname{tg} \frac{x}{2} - x + C$$

8.5.42 $\int \operatorname{tg}^7 x dx = [c.n. 6. \operatorname{tg}^2 = \frac{1}{\cos^2 x} - 1] =$

$$= \int \operatorname{tg}^5 x \cdot (\frac{1}{\cos^2 x} - 1) dx = \int \operatorname{tg}^5 x \cdot \frac{dx}{\cos^2 x} - \int \operatorname{tg}^5 x dx =$$

$$= [t = \operatorname{tg} x \Rightarrow dt = \frac{dx}{\cos^2 x}] = \int t^5 dt - \int \operatorname{tg}^5 x \cdot (\frac{1}{\cos^2 x} - 1) dx =$$

$$= \int t^5 dt - \int \operatorname{tg}^3 x \cdot \frac{dx}{\cos^2 x} - \int \operatorname{tg}^3 x dx =$$

$$= \int t^5 dt - \int t^3 dt - \int \operatorname{tg} x \cdot \left(\frac{1}{\cos^2 x} - 1 \right) dx =$$

$$= \int t^5 dt - \int t^3 dt - \int \operatorname{tg} x \cdot \frac{dx}{\cos^2 x} - \int \operatorname{tg} x dx =$$

$$= \int t^5 dt - \int t^3 dt - \int t dt - \int \frac{\sin x dx}{\cos x} = [y = \cos x \Rightarrow dy = -\sin x dx]$$

$$= \frac{t^6}{6} - \frac{t^4}{4} - \frac{t^2}{2} + \int \frac{dy}{y} + C =$$

$$= \frac{t^6}{6} - \frac{t^4}{4} - \frac{t^2}{2} + \ln|y| + C =$$

$$= \frac{\operatorname{tg}^6 x}{6} - \frac{\operatorname{tg}^4 x}{4} - \frac{\operatorname{tg}^2 x}{2} + \ln|\cos x| + C$$